

Shaharior Areej Pranto

+49 157 517 84208 | shahariorpranto@gmail.com | linkedin.com/in/shaharior-pranto | shahariorpranto.com | ORCID: 0009-0002-4916-8181

RESEARCH FOCUS

I am seeking a funded doctoral position in construction management. My core interest is distributed decision-making in construction project teams, built on ten years moving between materials research and organisational research, and informed by my MEng thesis finding that teams which spread decision-making capacity across levels were more resilient under compounding uncertainty than teams run by strong individual leaders. Around that spine sit genuine secondary interests: leadership and organisational behaviour, risk and crisis management, infrastructure delivery, procurement and supply-chain coordination, housing and project governance, and sustainable and circular construction implementation.

My methodological grounding is in qualitative and mixed-methods empirical research: semi-structured expert interviewing, survey design, manual thematic analysis, and multiple case study design. I am open to international relocation for the right funded position.

SKILLS

- **Research methods:** semi-structured expert interviewing, participant observation, life-history mapping, public-life study, bilingual survey design, reflexive thematic analysis (Braun and Clarke 2006), multiple case study design (Yin 2018), purposive sampling, triangulation, mixed-methods integration, controlled experimental design.
- **Tools:** Microsoft Excel for survey data; manual qualitative coding to date. Engineering software: Microsoft Project, AutoCAD, Revit Architecture, STAAD.Pro.
- **Engineering and testing:** geotechnical investigation, structural building evaluation, concrete and soil testing (compressive and flexural strength, permeability, thermal conductivity, Proctor compaction, CBR).
- **Languages:** Bangla (native); English (professional working proficiency); Hindi (intermediate); German (A1).

EDUCATION

MEng International Project Management, Building, Real Estate and Infrastructure

HFT Stuttgart (Hochschule für Technik Stuttgart), Germany | 2021 to 2024

- Overall grade 1.9; thesis grade 1.7 (German scale, where 1.0 is the highest achievable grade).
- Thesis: Leading Through Crisis: Identifying and Cultivating Leadership Skills in Construction. Sole-authored.
- Mixed-methods design: structured literature review; bilingual workforce survey (350 practitioners approached, 92 responses); nine semi-structured interviews using a 32-question protocol; manual thematic analysis (Braun and Clarke 2006) across two coding passes.
- Key finding: teams that spread decision-making capacity across levels proved more resilient under compounding uncertainty than teams run by strong individual leaders.
- Supervised by Prof. Jürgen Marc Volm and Prof. Dr. Uta Bronner.
- Coursework: Sustainability and Green Building; Research Methods and Strategic Decision-Making; Leadership and Change Management; Construction Economics; Risk Management; Contract Law.

BEng (Hons) Civil Engineering

Caledonian College of Engineering, Oman (affiliated with Glasgow Caledonian University, Scotland) | 2014 to 2018

- Second Class Honours, Upper Division; CGPA 3.09 out of 4.0. Runner-up, Star Student Award.
- Thesis: A Comparative Study of Compaction Energy on Lime and GGBS Stabilised Soil (see Research Experience).

RESEARCH EXPERIENCE

Research Assistant and Data Analyst (part-time) | MCTspaceLab, Muscat, Oman | Fieldwork in Oman and Lisbon, Portugal
In Oman:

- Supported qualitative research on the livelihood strategies of low-income Bangladeshi migrant workers.
- Conducted exploratory and in-depth interviews, later used for life-history mapping, alongside participant observation in workers' everyday environments.
- Carried out public-life studies, mapping activity and pedestrian counts in markets and public spaces, with photographic and sound documentation.

In Lisbon, Portugal:

- Supporting research with second-time migrants who relocated from Gulf countries, through exploratory and semi-structured interviews and participant observation.

Master's Thesis Researcher | HFT Stuttgart, Germany

- Designed and ran the full research programme independently, from literature synthesis and survey design through ethics approval and recruitment.

- Ran a bilingual workforce survey (350 practitioners approached, 92 responses) and nine semi-structured interviews with project managers, a CEO and a senior site engineer, using a 32-question protocol developed for this study.
- Coded every transcript by hand using thematic analysis, across two full coding passes.
- Synthesised survey and interview findings into a sole-authored thesis, graded 1.7.

Bachelor's Thesis Researcher | Caledonian College of Engineering, Oman

- Thesis: A Comparative Study of Compaction Energy on Lime and GGBS Stabilised Soil. Sole-authored.
- Designed a comparative experimental study, sourcing GGBS from Sohar Steel Industry and testing it as a partial lime substitute across four substitution ratios under standard and modified Proctor compaction.
- Compared compaction and bearing-capacity results under controlled, repeatable test conditions, with numerical measurement recorded across each ratio and compaction energy.
- Found that GGBS at 6 to 9 per cent substitution matched or beat lime's bearing capacity on its own, with the added benefit of diverting industrial by-product from landfill.

Team Member, FURAP Research Project | The Research Council of Oman

- Six-member undergraduate team, competitive grant of OMR 4,000, investigating PET waste as a fine aggregate replacement in concrete.
- My contribution centred on materials testing across compressive strength, flexural strength, thermal conductivity, and permeability, results analysis, and research and academic writing for the project report.
- The team found that concrete with up to 32 per cent PET replacement still reached C35-grade structural performance. This was the first funded research project of my career.

PUBLICATIONS AND RESEARCH OUTPUTS

Conference

- Annisa, S. E. and Pranto, S. A. Conference abstract, Deliver Safe and Social Housing (DASH) Conference, Belgrade, Serbia. Accepted; second author and presenting author.

Sole-authored pre-prints

- Leadership Skills for Crisis Management in the German Construction Industry: A Mixed-methods Study. SSRN pre-print, doi.org/10.2139/ssrn.7241638.
- Crisis Management Strategies and Organizational Challenges in the Construction Industry: A Structured Review with Practitioner Insights. SSRN pre-print, doi.org/10.2139/ssrn.7242178.
- Developing Crisis Leadership in Construction: The Role of Structured Training, Reflection, and Organizational Learning. SSRN pre-print, doi.org/10.2139/ssrn.7242619.
- Leadership Skills for Crisis Management in the Construction Industry: A Self-, Business- and People-Management Categorization. Zenodo pre-print, doi.org/10.5281/zenodo.21892978.
- Comparative Effect of Compaction Energy on Lime- and GGBS-Stabilised Fine Soil: A Laboratory Investigation. SSRN pre-print, doi.org/10.2139/ssrn.7242118.
- Effect of Lime and Ground Granulated Blast Furnace Slag on the Compaction Behaviour of Fine Soil: Full Experimental Report. Zenodo pre-print, doi.org/10.5281/zenodo.21893706.

Under review

- Manuscript under peer review with a Springer Nature journal.

Theses and research reports

- Leading Through Crisis: Identifying and Cultivating Leadership Skills in Construction. MEng thesis (unpublished), sole-authored, HFT Stuttgart.
- Performance Analysis of Eco-Concrete with PET as Fine Aggregate Replacement. Undergraduate team research report, The Research Council of Oman, FURAP Call 3. Six-member team; my contribution: materials testing, results analysis, and research and academic writing.
- A Comparative Study of Compaction Energy on Lime and GGBS Stabilised Soil. BEng thesis (unpublished), sole-authored, Caledonian College of Engineering.

INDEPENDENT RESEARCH INTERESTS

Ongoing, self-directed inquiries arising from professional practice; not published or peer-reviewed.

- **Client decision-making in residential interior design:** patterns in how clients across the ZAP portfolio weigh trade-offs and express aesthetic and budget preferences.
- **Client psychology in bespoke residential delivery:** how an individual client's motivations, priorities and anxieties unfold over a single project.
- **Direct-access market structures in Bangladesh:** how direct client access to wholesalers reshapes the roles of retailers and intermediaries in the construction materials market.

TEACHING EXPERIENCE

Industrial Supervisor, University Engineering Placements | Muscat Laboratory Company, Oman | 2018 to 2020

- Supervised undergraduate engineering students from several Omani universities during their mandatory industrial placements.
- Walked students through materials testing protocols, geotechnical investigation, and laboratory discipline, often for the first time in their working lives.
- Taught technical report writing and research proposal structure alongside the hands-on supervision.
- Gave individual written feedback on students' technical work.

Invited Guest Lecturer | Multiple institutions, Oman

- Taught Soil Mechanics, Building Evaluation, Material Testing, Sustainable Construction, and Academic Writing at Caledonian College of Engineering.
- Covered Construction Principles and Materials Engineering at Sultan Qaboos University.
- Delivered Material Testing and Building Evaluation sessions at the Ministry of Defence Engineering Department.

Private University Tutor | Oman, Bangladesh, and online | 2016 to 2025

- Provided paid academic support to university students.
- Covered soil mechanics, construction technology, materials testing, presentation skills, and research methodology.
- Adjusted my approach for students coming from very different academic backgrounds and confidence levels.

TEACHING PHILOSOPHY

I bring students into the problem rather than lecturing at them, using real project examples, including ones that failed, because those are what students remember. In placement supervision I have watched hesitant first-time report writers learn to present findings confidently to a laboratory client. That shift, watching someone's confidence catch up with their competence, is what I work towards in every teaching setting, and it is why I give detailed written feedback on technical work rather than a single grade.

TEACHING SUBJECTS

- **Core Teaching Areas:** Construction Management | Project Management | Soil Mechanics | Materials Testing and Technology | Geotechnical Investigation | Technical Report Writing | Research Methods | Sustainable Construction.

PROFESSIONAL EXPERIENCE

Operations Lead | ZAP Comfortable Homes, Bangladesh | 2020 to present

- Lead operations for a residential and commercial design-build firm with a 32-member team, with responsibility for procurement, contractor management, quality, risk and client relationships.
- Delivered more than 50 projects, running up to ten concurrent active sites at the busiest points.
- Negotiated 17 supplier contracts, bringing procurement cost down 25 to 30 per cent, and delivered the flagship residential project 60 days ahead of programme.

Student Assistant | IUSD, Universität Stuttgart, Germany | Feb 2022 to Aug 2024

- Coordinated international academic events, including thesis defences, workshops, and international field trips, from planning through to delivery.
- Owned end-to-end international student intake across a wide range of nationalities, from application processing through interview panel coordination.

Civil Engineer and Marketing Coordinator | Muscat Laboratory Company, Oman | 2018 to 2020

- Produced geotechnical investigation reports for government and private clients, covering borehole and trial-pit programmes, laboratory testing and foundation recommendations, to tight client turnaround times.
- Led geotechnical investigation on the Muscat International Airport Expansion, a high-pressure, multi-stakeholder infrastructure programme.
- Site engineering, site investigation and materials testing on the Al Amerat Road Expansion, with regular Ministry of Transport client briefings.
- Carried out structural building evaluations for commercial and residential buildings in line with regulatory requirements.

Earlier Engineering Roles and Training | Oman and Egypt | 2014 to 2017

- Engineering internships and site training alongside the BEng with Al Mazaya Trading and Contracting, Al Watanyiah United Engineering and Contracting, Modern Light Trading and Contracting and Technibuild Contracting in Oman, and Arabisk Construction Company in Cairo, Egypt.

AWARDS, CERTIFICATION AND MEMBERSHIPS

- Winner, British Council Public Speaking Competition, Oman; Runner-up, Star Student Award, Caledonian College of Engineering.

- Nominated Delegate, People to People World Leadership Forum, Washington, D.C.; Former President, AIESEC Oman, leading a team of 25.
- Certified Business Analyst Professional (CBAP), Qatar Advanced Technology and Research Centre; software training, CADD Centre, Oman.
- Individual Member, Association of Researchers in Construction Management (ARCOM).
- Member and Volunteer, EZBET e.V., Germany; Former Member, Chartered Institute of Building (CIOB), United Kingdom.

SELECTED OPEN RESEARCH QUESTIONS

- How does distributed decision-making capacity develop and sustain itself in teams facing circular renovation and design-for-disassembly requirements?
- What supervisory conditions let information and permission to act reach every level of a team, rather than staying concentrated in leadership?
- How do industrial by-products like GGBS and PET move from validated lab findings to default specification on live projects, and what actually stops them?

REFEREES

Four academic and professional referees, including both MEng thesis supervisors, are available on request.